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ORIGINAL PAPER

Sound production mechanism in the Brazilian spiny lobsters (Family Palinuridae)

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Abstract
Strident lobsters of the Palinuridae family emit sounds in the presence of predators that can be used in the acoustic monitoring of such species. This study aims to identify sound emission and describe the structures of sound mechanism in red (*Panulirus meripurpuratus*) and green (*Panulirus laevicauda*) Brazilian spiny lobsters. The lobsters were collected in the environment, and recordings were performed in laboratory tanks with submerged hydrophones. The animals were stimulated to emit sound by the presence of an artificial octopus in the tank. The sounds emitted by the lobsters were analyzed by means of the waveform, and the structures involved (plectrum and file) were observed using scanning electron microscopy (SEM). Both species emitted a rasp sound composed of several pulses, with a rate varying from 125 to 265 pulses per second. The SEM showed that the file is similar between the species and it is covered by microscopic plates (shingles) measuring between 12.6 and 12.9 µm of medial–lateral width and 6.2 µm and 7.1 µm of anteroposterior length. This analysis also revealed for the first time the presence of pores (< 1 µm in diameter) with vestiges of cuticular setae in-between the shingles, which could be involved in the mechanosensory control of sound production in lobsters. The results of this study corroborate the morphological pattern of sound emission structure described for other lobsters of the genus *Panulirus*.

Keywords *Panulirus* · Stridulation · Bioacoustics · Rasp

Introduction

Several species of crustaceans are capable of producing sounds using different emission mechanisms (Schmitz 2002). These sounds compound the soundscape of marine environments (Au and Hastings 2008; Coquereau et al. 2016a) and could provide important information on the occurrence, distribution and behavior of species.

Crustaceans can emit sounds associated with different situations, such as during feeding activity of penaeid shrimps by the shock between mandibles (Smith and Tabrett 2013; Silva et al. 2019) or territorialistic behavior of snapping shrimps (*Alpheus* and *Synalpheus*) by quickly closing their chela for sound production (Au and Banks 1998; Lillis et al. 2017). Additionally, lobsters of the genus *Homarus*, when threatened, emit a buzzing sound by the vibration of their shells during the contraction of muscles located at the base of the second antenna (Henninger and Watson 2005; Jézéquel et al. 2018).

Another common sound in crustaceans is the stridulation, which is produced by the friction of two parts of the body and it is generally associated with four types of behavior: sexual, territorial, social and intimidation (Guinot-Dumortier and Dumortier 1960; Popper et al. 2001). In hermit crabs of the genus *Trizopagurus*, sound is generated by the stridulatory system located in the chela and it is related to a defense behavior (Field et al. 1987). In lobsters of the Palinuridae family, sound pulses (called rasps) are also produced by a strident organ (Parker 1878). In this case, the friction mechanism is called "stick-and-slip", analogous to

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what occurs with a bow and the strings of a musical instrument (Patek 2001).

The stridulatory system in lobsters of the Palinuridae family is formed by the plectrum, located on the inner surface of the antenna base and composed of ridges of soft tissue, and the file, located below the eyes on the antennular plate and formed by microscopic plates, called "shingles" (Moulton 1957; Patek 2001; Patek and Oakley 2003). The sound is produced only when the plectrum moves in the posterior direction and is rubbed against the anteriorly projected shingles on the file (Patek 2002). The sound produced by such lobsters is commonly associated with a defense behavior against predators, such as octopus, or an alert for the conspecifics (Cobb 1981; Bouwma and Herrnkind 2009; Patek et al. 2009; Buscaino et al. 2011).

The red (*Panulirus meripurpuratus* Giraldes and Smyth 2016) and green (*Panulirus laevicauda* Latreille, 1817) lobsters are important fishing resources in Brazil (Lima and Andrade 2017). Their estimated production is 6100 t (FAO 2018), which corresponds to 27.6% of Brazil's total seafood export value (MDIC 2019). However, in order to promote the recovery of stocks and the preservation of lobster species, the Brazilian Institute for the Environment and Renewable Natural Resources established that fishery must be suspended from December to May (IBAMA 2008). The *P. meripurpuratus*, until recently known as *Panulirus argus* (Latreille, 1804), is a new species, as they were separated into genetically distinct populations isolated by the Amazon River (Giraldes and Smyth 2016).

Passive acoustic monitoring provides important information about biodiversity, health and temporal changes in coral reefs and other marine environments (Lammers and Munger 2016) and can be used as a tool to monitor populations of economic interest, such as lobsters (Kikuchi et al. 2015). Therefore, the characterization of the acoustic signatures of different species is fundamental to their identification in the soundscape of aquatic environments (Coquereau et al. 2016b).

Although the emission of sound in lobsters of the Palinuridae family has been previously described, there is no information available in the literature for the Brazilian species. Thus, the study aims to identify sound emission and describe the structures and sound mechanism associated with red and green lobsters from the northeastern coast of Brazil.

Materials and methods

Collection and maintenance of lobsters in the laboratory

Four lobsters were manually collected at depth of 25 m on the coast of Recife, Pernambuco, Brazil (08°10′S and 34°49′W). The lobsters were transported in seawater with constant aeration to the laboratory at the Universidade Federal Rural de Pernambuco, where two were identified as *P. laevicauda* and the two others as *P. meripurpuratus* (Giraldes and Smyth 2016). The individuals were separated by species and maintained in two circular fiberglass tanks (500 L; 1 m in diameter) connected to a water recirculation system with mechanical and biological filters, salinity of 35, temperature of 27.2 ± 0.7 °C and dissolved oxygen of 6.9 ± 0.6 mg/L. At the bottom of each tank, two pieces of PVC tube (10 cm in diameter; 25 cm in length) were placed to serve as shelter for the animals.

The animals were fed with hake fillet (*Merluccius hubbsi*) and shellfish (*Anomalocardia flexuosa*) at a rate of 10% of the biomass per day, supplied once a day at 16:00. Uneaten food and feces were removed from the tank in the next morning.

Sound acquisition and analysis

For the acquisition and characterization of the sound, a specimen of *P. laevicauda* and *P. meripurpuratus* was selected 26 days after the ecdysis, with carapace length of 53.1 mm and 53.2 mm and total weight of 139.2 g and 143.9 g, respectively. The recordings were performed with each species separately, in circular tanks (500 L; 1 m in diameter) with water temperature of 25.7 ± 0.1 °C and salinity of 35, where the lobsters were stimulated to emit sound by simulating an attack with an artificial octopus (total length = 21 cm) in the tanks.

The sound was recorded with an omnidirectional hydrophone (SoundTrap 300 STD, Ocean InstrumentsNZ) with linear frequency response from 2 Hz to 60 kHz, sampling rate of 288 kS/sec (16 bits) and sensitivity of -173 dB, positioned in the center of the tank at 20 cm from the bottom. This hydrophone has an anti-aliasing filter set to 60 kHz, resulting in a 12-dB roll-off per octave. Four recordings of 30 min for each species were analyzed using the Raven® 1.5 Pro software (Cornell Laboratory of Ornithology, Ithaca, New York) for the manual identification and counts of the sound rasps of the recordings plotted on an oscillogram and spectrogram (type: hann; overlap: 50%; resolution: 512 sample window size; 3 dB filter bandwidth at 809 Hz resolution). After identification, 20 rasps for *P. laevicauda* and 12 rasps for *P. meripurpuratus* were randomly selected and analyzed for the acoustic characterization of each species [rasp duration (ms), number of pulses per rasp and pulse rate (number of pulses per rasp divided by rasp duration)]. The rasp duration was defined as the time interval between the arrival of 5 and 95% of the total rasp energy (Blackwell et al. 2004).

Analysis of the structures responsible for sound emission

After the recordings, the animals were desensitized and slaughtered with water and ice, while their structure of sound production (file and plectrum) was dissected with the aid of a scalpel, scissors and pincers. The right file and plectrum were then examined under a stereoscopic microscope (Coleman/NSZ-405-T-LED), to which a camera (Touptek/UCMOS03100KPA) was coupled to capture the images using the Toupview software version 3.2 (ToupTek Photonics).

The left file and plectrum were sectioned into pieces of approximately 1 cm and fixed in 2.5% glutaraldehyde in 0.1 M cacodylate buffer, pH 7.2. These pieces were post-fixed for 30 min in 1% OsO_4, dehydrated in ethanol and critical point dried with liquid CO_2. The dried material was coated with gold–palladium to a thickness of 15 nm and then observed with a Jeol JSM-5600 scanning electron microscope (SEM), operating at 15 kV.

Ten images of the central portion of the file of each species (1400 × magnification; 88 × 66 µm field of view-640 × 480 pixels) were analyzed to count the total number of shingles per field of view, and to measure the greatest medial–lateral and anterior–posterior distances (µm) of 100 shingles of each species. Measurements were performed after calibrating the pixel size according to the real image scale using the software ImageTool version 3.0 (University of Texas Health Science Center, San Antonio, Texas).

Results

The two species studied emitted a rasp formed by several pulses (Fig. 1). *Panulirus laevicauda* showed a mean rasp duration of 52 ± 18 ms and *P. meripurpuratus* of 38 ± 12 ms. The mean number of pulses per rasp was 8.9 ± 2.8 for *P. laevicauda* and 7.5 ± 3.0 for *P. meripurpuratus*, while the mean of pulse rate was 173.9 ± 20.3 and 194.8 ± 38.2 for *P. laevicauda* and *P. meripurpuratus*, respectively. Table 1

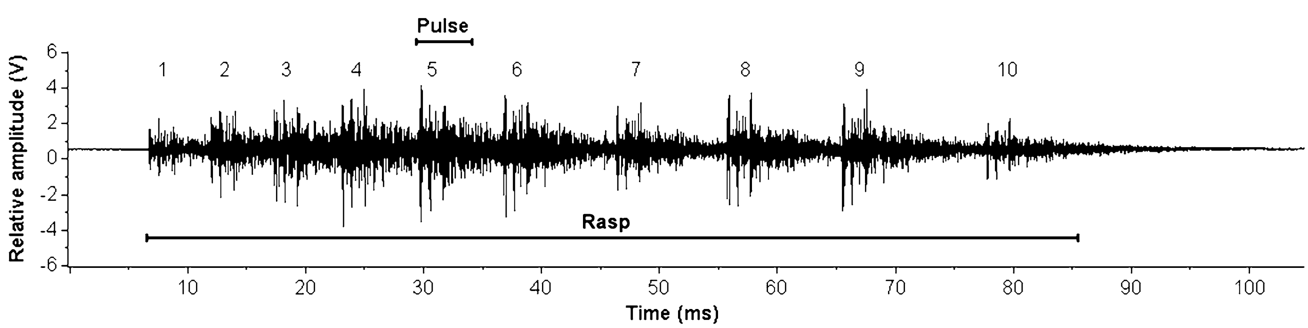


Fig. 1 Oscillogram of a rasp produced by the green lobster (*Panulirus laevicauda*) composed of ten pulses (numbers from 1 to 10) and duration of 78 ms

Table 1 Descriptive statistics for rasp features reported in the literature for spiny lobster of the genus *Panulirus* (family Palinuridae). [Adapted from Patek and Oakley (2003)]

Species	*N*	#rasps	Rasp duration (ms)			Number of pulses per rasp			Pulse rate (pulses per second)			References
			Mean (SD)	Min	Max	Mean (SD)	Min	Max	Mean (SD)	Min	Max	
P. argus	4	43–44	151.7 (82.3)	–	448.8	7.2 (1.5)	–	21	65.9 (27.4)	–	147.8	Patek and Oakley (2003)
P. guttatus	1	9	53.3	26	72	–	–	–	–	–	–	Hazlett and Winn (1962)
P. homarus	1	17–21	89.5 (83.0)	–	262.2	9.9 (8.6)	–	30	120.6 (19.8)	–	149.3	Patek and Oakley (2003)
P. interruptus	19	5–21	108 (35)	15	303	7 (3)	2	19	71 (20)	24	192	Patek et al. (2009)
P. japonicus	1	11	155.1 (51.6)	–	221.8	9.5 (5.0)	–	23	61.4 (21.0)	–	103.7	Patek and Oakley (2003)
P. laevicauda	**1**	**20**	**52 (18)**	**24**	**86**	**8.9 (2.8)**	**5**	**15**	**173.9 (20.3)**	**139.5**	**208.3**	**Present study**
P. longipes	1	11–17	101.4 (66.5)	–	260.7	11.4 (7.8)	–	29	147.4 (17.0)	–	174.8	Patek and Oakley (2003)
P. meripurpuratus	**1**	**12**	**38 (12)**	**15**	**58**	**7.5 (3.0)**	**3**	**13**	**194.8 (38.2)**	**125.0**	**265.3**	**Present study**

N number of animals, *#rasps* number of rasps analyzed, *SD* standard deviation, *Min* minimum, *Max* maximum

Data for *P. laevicauda* and *P. meripurpuratus* was recorded in the present study (in bold type) presents the summarized sound data for the green and red lobster, as well as for other species for comparison purposes.

Stereoscopic microscope observations showed similar characteristics of the plectrum, as well as the file, for *P. laevicauda* and *P. meripurpuratus* (Fig. 2). The plectrum is located above the flap on the inner surface of the antenna base and composed by ridges of soft-textured tissue. In the plectrum of both species, the presence of a knob is observed (Fig. 2a, b), which is a protuberance that fits in the groove located in the lateral portion of the file (Fig. 2c, d), allowing the plectrum to slide in alignment.

SEM of plectrum and file also revealed similarities between the two species (Figs. 3, 4). The plectrum ridges are fused together in the anterior region (Fig. 3a, b) and separated in the intermediate portion (Fig. 3c, d) and posterior (Fig. 3e, f). The file is covered by microscopic plates or shingles (Fig. 4a–d), slightly protruding in their anterior part, with a small elevation in the central region (Fig. 4e, f). The shingles of *P. laevicauda* measured 12.9 ± 1.2 µm of medial–lateral width and 6.2 ± 0.6 µm of anterior–posterior length, while for *P. meripurpuratus* these values were 12.6 ± 1.4 µm and 7.1 ± 0.7 µm, respectively. The total number of shingles per field of view was similar between the two species, varying from 68.5 ± 4.1 (*P. laevicauda*) to 67.0 ± 8.6 (*P. meripurpuratus*).

The presence of pores in-between the shingles was also evidenced by SEM, following a spaced pattern arrangement along the file in both species (Fig. 4a, b). These pores have a diameter of approximately 0.8 µm, and for some of them it was possible to observe vestiges of a sensory cuticular seta (Fig. 4g, h).

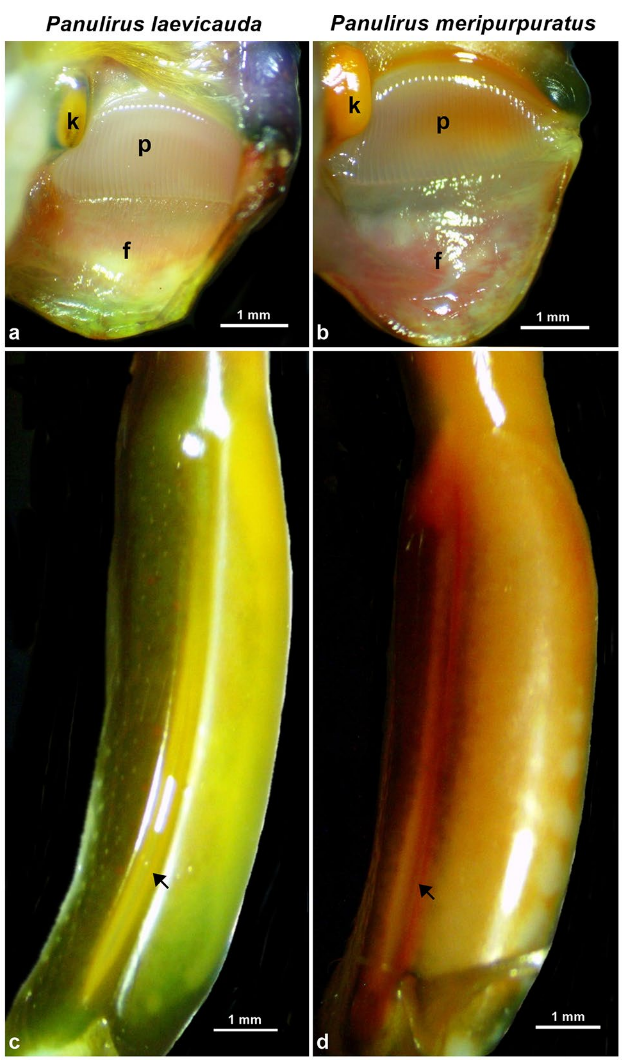


Fig. 2 Right plectrum and file of *Panulirus laevicauda* (**a**, **c**) and *Panulirus meripurpuratus* (**b**, **d**). The plectrum (p) is located above the flap (f) on the inner surface of the posterior antenna base and consists of ridges of soft tissue. The knob (k) is a protuberance located on the lateral side of the plectrum (**a**, **b**). On the lateral side of the file is a groove (arrow) where the knob fits (**c**, **d**)

Discussion

The green (*P. laevicauda*) and red (*P. meripurpuratus*) lobsters emit a rasp sound, composed of several pulses, as described for other lobsters of the Palinuridae family (Moulton 1957; Meyer-Rochow and Penrose 1976; Patek 2001; Patek and Oakley 2003; Patek et al. 2009). Generally, this sound emission is associated with defense behavior against predators, such as octopus (Bouwma and Herrnkind 2009; Patek et al. 2009; Buscaino et al. 2011), allowing lobsters to discourage an imminent attack (Patek 2001). This behavior was observed in the present study using an artificial octopus, suggesting that this is a viable approach that does not require the capture and maintenance of another living organism to act as a predator in acoustic research with strident lobsters in captivity.

In comparison with other lobsters of the genus *Panulirus*, both species of this study presented a similar rasp duration to that found in *Panulirus guttatus* (Latreille, 1804) (Hazlett and Winn 1962), but lower (2–3 times) than the other species (Table 1). Additionally, the average number of pulses per rasp was similar in all species of the genus. Although the pulse rate shows a positive correlation with the size of the animal, the characteristics of the rasps may be more influenced by the movement of one or two antennas (Patek et al. 2009). This behavior is probably associated with the animals ability to control the stick-and-slip mechanism of sound production (Patek and Baio 2007). In the present study, video images were not obtained concomitantly with the sound recording, making it impossible to relate the characteristics of the rasps with the influence of the movement of only one antenna or both simultaneously or sequentially, and their relationship with the behavior of lobsters.

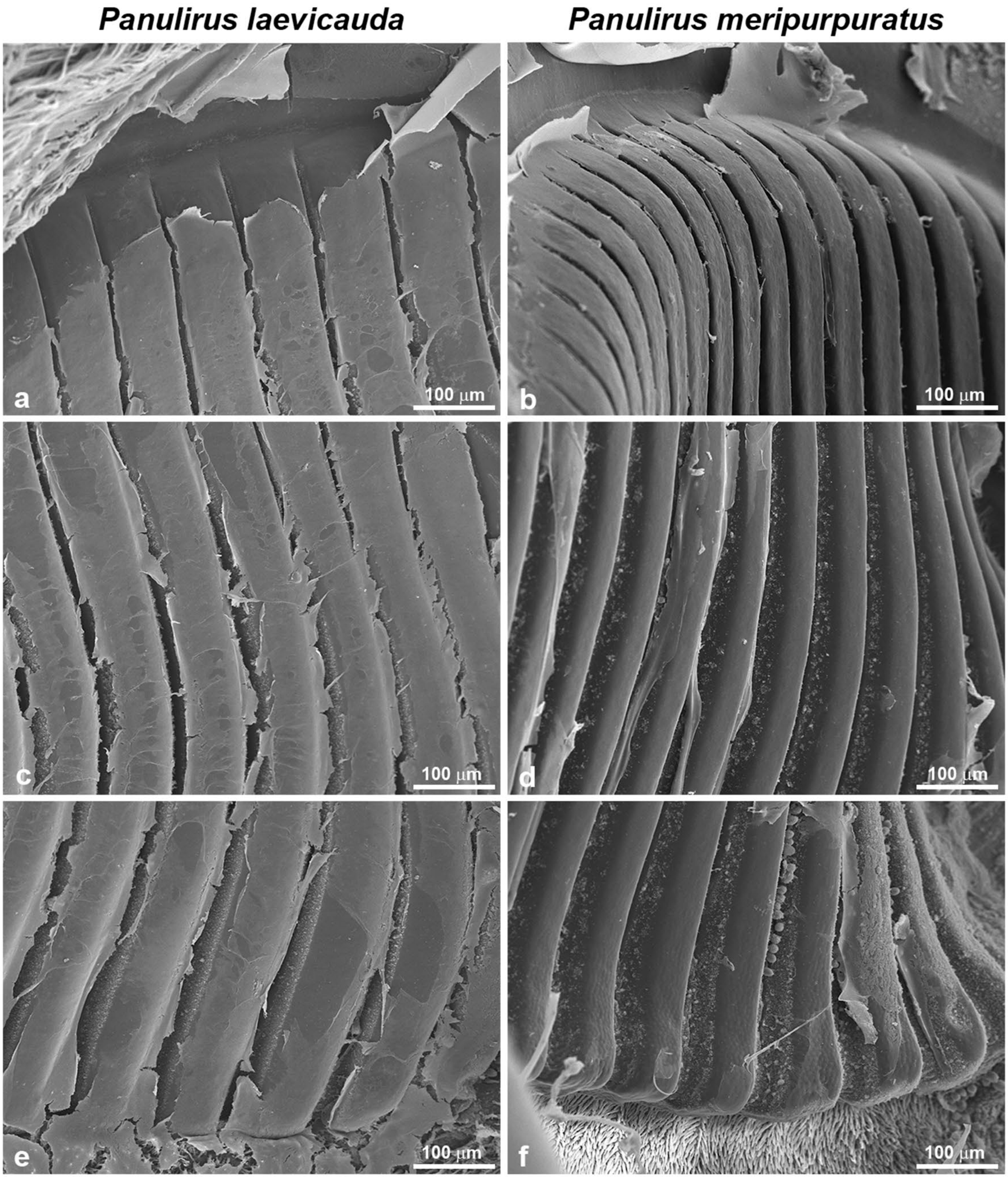
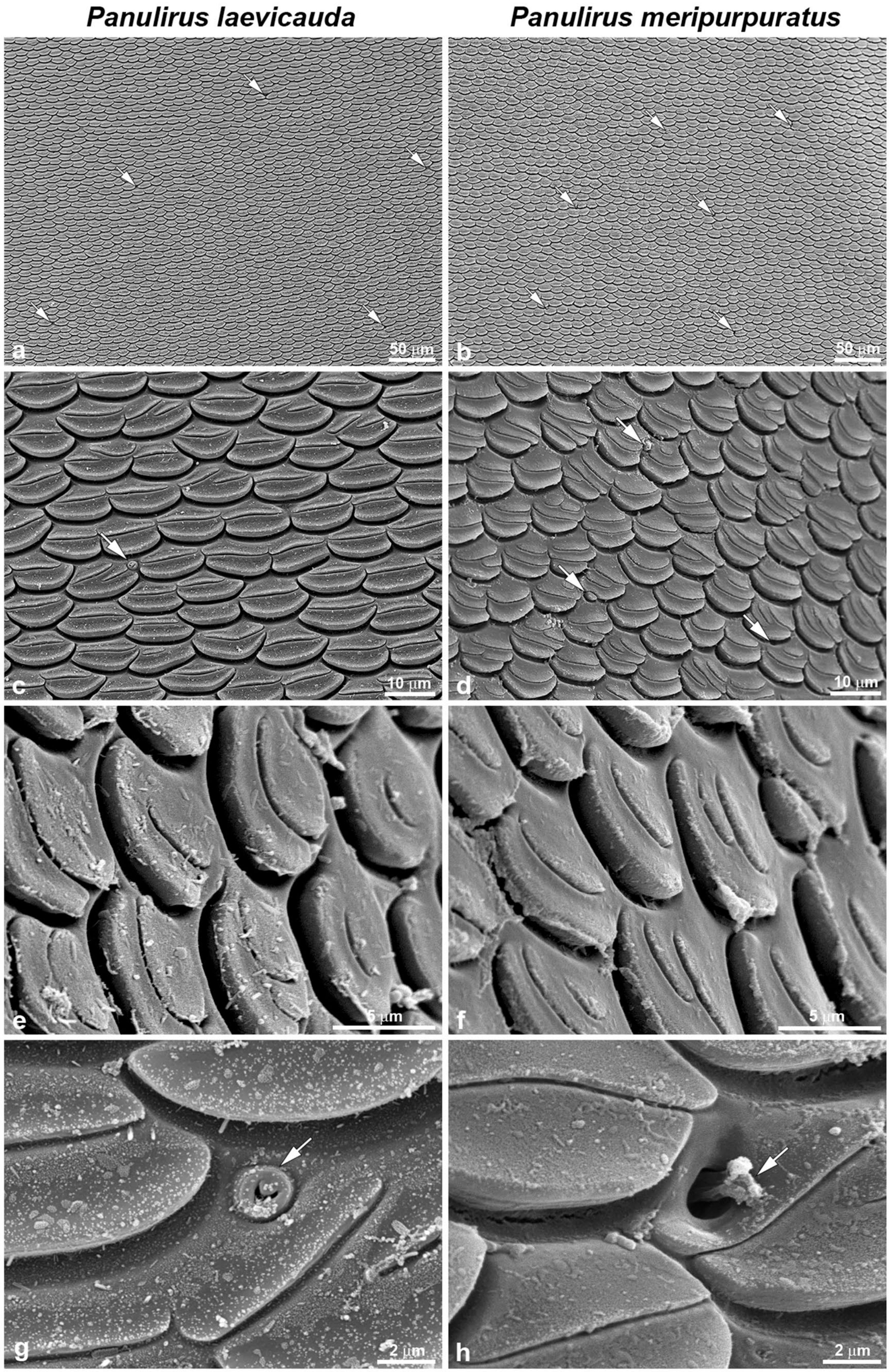


Fig. 3 Scanning electron micrographs of plectrum from *Panulirus laevicauda* (**a**, **c**, **e**) and *Panulirus meripurpuratus* (**b**, **d**, **f**). The anterior edge of the plectrum with ridges attaches to the ventral surface of the antennal base (**a**, **b**). The intermediate (**c**, **d**) and posterior portion of the ridges that slide over the shingles (**e**, **f**)

Green and red lobsters have a sound emission mechanism similar to that described for other species of the genus *Panulirus* (Moulton 1957; Meyer-Rochow and Penrose 1976; Patek 2002; Patek and Baio 2007). The file is located on the antennular plate and is covered by shingles with their prominent anterior portion, allowing greater friction with the plectrum in strident lobsters (Patek 2002; Patek and Baio 2007). Both species also present a small elevation in the central region of the shingles as reported for *P. argus* (Patek 2002). In *Panulirus longipes* (A. Milne-Edwards, 1868), the origin of this additional ridge could be the fuse of two or three units of shingles during early development phases (Meyer-Rochow and Penrose 1976). This characteristic of the shingle may affect the friction of the plectrum on the file and, consequently, the stick-and-slip mechanism of sound emission (Patek 2002).


Panulirus laevicauda
Panulirus meripurpuratus
a
50 µm
b
50 µm
c
10 µm
d
10 µm
e
5 µm
f
5 µm
g
2 µm
h
2 µm

◂**Fig. 4** Scanning electron micrographs of file from *Panulirus laevicauda* (**a**, **c**, **e**, **g**) and *Panulirus meripurpuratus* (**b**, **d**, **f**, **h**). The file surface is covered with shingles that are slightly protruding in their anterior portion (to the bottom of the page) with a small elevation in the medial region (**c**, **d**). An oblique image of the shingles evidenced the protruding leading edges and the additional ridge in the medial region (**e**, **f**). Arrows indicate the pores in-between the shingles (**a**–**d**, **g**, **h**), with vestiges of a sensory cuticular seta (**h**)

Furthermore, the size of the shingles of *P. laevicauda* and *P. meripurpuratus* was similar to that observed for *Panulirus interruptus* (Randall, 1840), as well as close to the width values (≈ 10 µm) recorded in other Palinuridae species (Patek and Baio 2007). The fact of the size and number of shingles per area were similar between both species may justify the similarity in the general characteristics of the rasps produced by green and red lobsters.

The present study was the first to identify the presence of pores located between the shingles along the file in Palinuridae lobsters. The function of such structures is not clear; however, in some of these pores vestiges were identified of what would probably be a sensory cuticular seta. Such pores are similar to those found in the antennas, from which emerge "mechanosensory hairs" that may act not only as mechanoreceptors to water movements (Tazaki and Ohnishi 1974), but also as chemoreceptors (Vedel 1985). These setae located in the file might be involved in the control of the sound production system, regulating the velocity of the plectrum and the force it exerts on the file, which are factors that influence the acoustic properties of the sound emitted by the stick-and-slip mechanism (Patek and Baio 2007). The control of sound production was observed in *P. longipes*, which often moved the antenna backward but without emitting sound (Meyer-Rochow and Penrose 1976). In addition, these sensory setae could also be involved in sound reception as described for crabs (Davie et al. 2015), but such hypotheses must be further evaluated in future studies.

In conclusion, our work confirmed that the lobsters *P. laevicauda* and *P. meripurpuratus* have a similar stridulating mechanism, such as other lobsters of the genus *Panulirus*. Furthermore, this similarity in the morphological pattern was reflected in the general characteristics of the rasps sound produced by these lobsters.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All applicable international, national and/or institutional guidelines for the care and use of animals were followed. Sample collections were carried out under the license of ICMBIO (SISBIO Permit 67042-1). This article does not contain any studies with human participants performed by any of the authors.

References

Au WWL, Banks K (1998) The acoustics of the snapping shrimp *Synalpheus parneomeris* in Kaneohe Bay. J Acoust Soc Am 103:41–47. https://doi.org/10.1121/1.423234

Au WWL, Hastings MC (2008) Principles of marine bioacoustics. Springer, New York. https://doi.org/10.1007/978-0-387-78365-9

Blackwell SB, Lawson JW, Williams MT (2004) Tolerance by ringed seals (*Phoca hispida*) to impact pipe-driving and construction sounds at an oil production island. J Acoust Soc Am 115:2346–2357. https://doi.org/10.1121/1.1701899

Bouwma PE, Herrnkind WF (2009) Sound production in Caribbean spiny lobster *Panulirus argus* and its role in escape during predatory attack by *Octopus briareus*. New Zeal J Mar Freshw Res 43:3–13. https://doi.org/10.1080/00288330909509977

Buscaino G, Filiciotto F, Gristina M, Buffa G, Bellante A, Maccarrone V, Patti B, Mazzola S (2011) Defensive strategies of European spiny lobster *Palinurus elephas* during predator attack. Mar Ecol Prog Ser 423:143–154. https://doi.org/10.3354/meps08957

Cobb JS (1981) Behaviour of the western Australian spiny lobster, *Panulirus cygnus* George, in the field and laboratory. Aust J Mar Freshw Res 32:399–409. https://doi.org/10.1071/MF9810399

Coquereau L, Grall J, Chauvaud L, Gervaise C, Clavier J, Jolivet A, Di Iorio L (2016a) Sound production and associated behaviours of benthic invertebrates from a coastal habitat in the north–east Atlantic. Mar Biol 163:127. https://doi.org/10.1007/s00227-016-2902-2

Coquereau L, Grall J, Clavier J, Jolivet A, Chauvaud L (2016b) Acoustic behaviours of large crustaceans in NE Atlantic coastal habitats. Aquat Biol 25:151–163. https://doi.org/10.3354/ab00665

Davie PJF, Guinot D, NG PKL (2015) Anatomy and functional morphology of Brachyura. In: Castro P, Davie PJF, Guinot D, Schram FR, Klein JCV (eds) Treatise on zoology-anatomy, taxonomy, biology. The crustacea, volume 9, part C–I: Decapoda: Brachyura. Brill, Leiden, pp 11–163

FAO (2018) Fishery and aquaculture statistics. Global capture production 1950–2016 (Fishstat J). In: FAO Fisheries and Aquaculture Department. Rome. www.fao.org/fishery/statistics/software/fishstatj/en. Accessed 16 Apr 2019

Field LH, Evans A, Macmillan DL (1987) Sound production and stridulatory structures in hermit crabs of the genus *Trizopagurus*. J Mar Biol Assoc UK 67:89–110. https://doi.org/10.1017/S0025315400026382

Giraldes BW, Smyth DM (2016) Recognizing *Panulirus meripurpuratus* sp. nov. (Decapoda: Palinuridae) in Brazil—systematic and biogeographic overview of *Panulirus* species in the Atlantic Ocean. Zootaxa 4107:353–366. https://doi.org/10.11646/zootaxa.4107.3.4

Guinot-Dumortier D, Dumortier B (1960) La stridulation chez les crabes. Crustaceana 1:117–155. https://doi.org/10.1163/156854060X00168

Hazlett A, Winn HE (1962) Sound production and associated behavior of bermuda crustaceans (*Panulirus*, *Gonodactylus*, *Alpheus*, and *Synalpheus*). Crustaceana 4:25–38. https://doi.org/10.1163/156854062X00030

Henninger HP, Watson WH (2005) Mechanisms underlying the production of carapace vibrations and associated waterborne sounds in the American lobster, *Homarus americanus*. J Exp Biol 208:3421–3429. https://doi.org/10.1242/jeb.01771

IBAMA (2008) Instrução Normativa do Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis n° 206, de 14 de novembro de 2008. http://www.icmbio.gov.br/cepsul/legislacao/instrucao-normativa/346-2008.html. Accessed 22 Apr 2019

Jézéquel Y, Bonnel J, Coston-Guarini J, Guarini JM, Chauvaud L (2018) Sound characterization of the European lobster *Homarus gammarus* in tanks. Aquat Biol 27:13–23. https://doi.org/10.3354/ab00692

Kikuchi M, Akamatsu T, Takase T (2015) Passive acoustic monitoring of Japanese spiny lobster stridulating sounds. Fish Sci 81:229–234. https://doi.org/10.1007/s12562-014-0835-6

Lammers MO, Munger LM (2016) From shrimp to whales: biological applications of passive acoustic monitoring on a remote Pacific coral reef. In: Au W, Lammers M (eds) Listening in the ocean. Modern acoustics and signal processing. Springer, New York, pp 61–81. https://doi.org/10.1007/978-1-4939-3176-7_4

Lillis A, Perelman JN, Panyi A, Mooney TA (2017) Sound production patterns of big-clawed snapping shrimp (*Alpheus spp.*) are influenced by time-of-day and social context. J Acoust Soc Am 142:3311–3320. https://doi.org/10.1121/1.5012751

Lima SAO, Andrade HA (2017) Variações nas capturas das lagostas vermelha (*Panulirus meripurpuratus*), verde (*Panulirus laevicauda* Latreille, 1817) e sapata (*Scyllarides brasiliensis* Rathbun, 1906) na costa de Pernambuco. Bol Inst Pesca 43:194–206. https://doi.org/10.20950/1678-2305.2017v43n2p194

MDIC (2019) ComexStat. http://comexstat.mdic.gov.br/pt/geral. Accessed 21 Feb 2019

Meyer-Rochow VB, Penrose JD (1976) Sound production by the western rock lobster *Panulirus longipes* (Milne Edwards). J Exp Mar Biol Ecol 23:191–209. https://doi.org/10.1016/0022-0981(76)90141-6

Moulton J (1957) Sound production in the spiny lobster *Panulirus argus* (Latreille). Biol Bull 113:286–295. https://doi.org/10.2307/1539086

Parker TJ (1878) Note on the stridulatting organ of *Panulirus vulgaris*. Proc Zool Soc Lond 1878:442–444

Patek SN (2001) Spiny lobsters stick and slip to make sound. Nature 411:153–154. https://doi.org/10.1038/35075656

Patek SN (2002) Squeaking with a sliding joint: mechanics and motor control of sound production in palinurid lobsters. J Exp Biol 205:2375–2385

Patek SN, Baio JE (2007) The acoustic mechanics of stick–slip friction in the California spiny lobster (*Panulirus interruptus*). J Exp Biol 210:3538–3546. https://doi.org/10.1242/jeb.009084

Patek SN, Oakley TH (2003) Comparative tests of evolutionary tradeoffs in a palinurid lobster acoustic system. Evolution 57:2082–2100. https://doi.org/10.1554/02-608

Patek SN, Shipp LE, Staaterman ER (2009) The acoustics and acoustic behavior of the California spiny lobster (*Panulirus interruptus*). J Acoust Soc Am 125:3434–3443. https://doi.org/10.1121/1.3097760

Popper AN, Salmon M, Horch KW (2001) Acoustic detection and communication by decapod crustaceans. J Comp Physiol A 187:83–89. https://doi.org/10.1007/s003590100184

Schmitz B (2002) Sound production in Crustacea with special reference to the Alpheidae. In: Wiese K (ed) The crustacean nervous system. Springer, Berlin, pp 536–547

Silva JF, Hamilton S, Rocha JV, Borie A, Travassos P, Soares R, Peixoto S (2019) Acoustic characterization of feeding activity of *Litopenaeus vannamei* in captivity. Aquaculture 501:76–81. https://doi.org/10.1016/j.aquaculture.2018.11.013

Smith DV, Tabrett S (2013) The use of passive acoustics to measure feed consumption by *Penaeus monodon* (giant tiger prawn) in cultured systems. Aquac Eng 57:38–47. https://doi.org/10.1016/j.aquaeng.2013.06.003

Tazaki K, Ohnishi M (1974) Responses from tactile receptors in the antenna of the spiny lobster *Panulirus japonicus*. Comp Biochem Phys A 47:1323–1327. https://doi.org/10.1016/0300-9629(74)90106-6

Vedel JP (1985) Cuticular mechanoreception in the antennal flagellum of the rock lobster *Palinurus vulgaris*. Comp Biochem Phys A 80:151–158. https://doi.org/10.1016/0300-9629(85)90532-8